ASSIGNMENT 1 ON GRAPH THEORY AS A MODELING TOOL IN DATA SCIENCE AND AI		
Student's Code		Deadline
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		Lecturer: [Lecturer Name]

Graph Theory as a Modeling Framework in Data Science and Artificial Intelligence

An Analytical Report

Introduction

As AI systems increasingly operate on interconnected, heterogeneous data, the limitations of flat tabular representations have become structurally apparent. Graph theory provides a mathematically principled language for encoding entities and their relationships, making it indispensable for problems where topology and connectivity are semantically meaningful. This report analyzes two domains—**Knowledge Graphs** for AI reasoning and **Fraud Detection** in financial networks—and examines why relational structure is often constitutive of the problem itself.

1 Domain I: Knowledge Graphs for AI Reasoning

1.1 The Real-World Problem

Large-scale question answering requires AI systems to integrate heterogeneous factual knowledge—biographical data, geographic relationships, scientific facts, and temporal dependencies. Systems such as Google's Knowledge Graph must support multi-hop queries: answering *"What language is spoken in the birthplace of the CEO of company X?"* demands chaining across at least three entity relationships.

1.2 Graph Reformulation

A knowledge graph (KG) is modeled as a directed multigraph $\mathcal{G} = (V, E, \mathcal{R})$, where V is a set of entities, $\mathcal{R}$ a set of relation types, and $E \subseteq V \times \mathcal{R} \times V$ typed directed edges (triples). As shown in Figure 1, the path $\textit{Einstein} \xrightarrow{\texttt{bornIn}} \textit{Ulm} \xrightarrow{\texttt{officialLanguage}} \textit{German}$ is a two-hop traversal. Multi-hop queries reduce directly to *path traversal* problems over this graph.

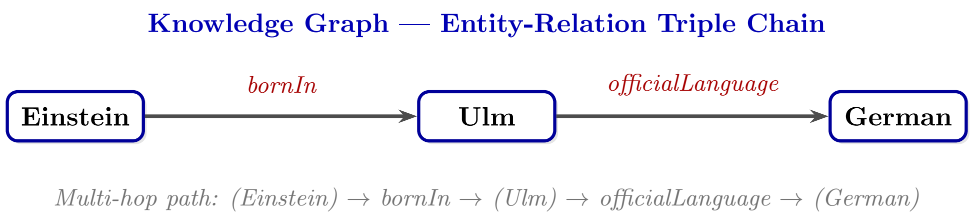

Figure 1: Knowledge graph triple chain: entity nodes connected by typed directed edges enable multi-hop reasoning.

1.3 Main Graph Concepts and Algorithms

TransE [1] learns low-dimensional embeddings interpreting relations as translations: $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ for valid triples, reducing link prediction to nearest-neighbor search. Path-ranking algorithms [4] rank reasoning paths via random walks. **Graph Attention Networks** [8] apply attention-weighted neighborhood aggregation, enabling context-sensitive entity disambiguation.

1.4 Why Classical Tabular ML Is Insufficient

Encoding all entity pairs requires $O(|V|^2)$ columns per relation type—untenable at Wikidata's 100 million entity scale. Multi-hop inference cannot be captured by any fixed feature set: the graph topology *is* the signal, not a byproduct of it.

1.5 Strengths and Limitations

Strengths. Each inference step corresponds to a traceable edge traversal. KG-augmented language models [6] show significant factual accuracy improvements. **Limitations.** KGs suffer from inherent incompleteness (open-world assumption), high construction cost, and the static-graph assumption [5].

2 Domain II: Graph-Based Fraud Detection in Financial Networks

2.1 The Real-World Problem

Financial fraud involves networks of colluding actors whose individual behaviors appear innocuous, but whose collective patterns reveal illicit structure. The signal lies in the *topology*: shared devices, overlapping beneficiaries, cyclic money flows, and synchronized behavioral patterns.

2.2 Graph Reformulation

The financial system is modeled as $\mathcal{G} = (V, E, X)$. As shown in Figure 2, V contains account, device, and merchant nodes; E are directed transaction edges; shared device links encode colluding infrastructure. The red directed cycle Acc A→B→C→D→A is the fraud ring signature—high internal density invisible to flat tabular models.

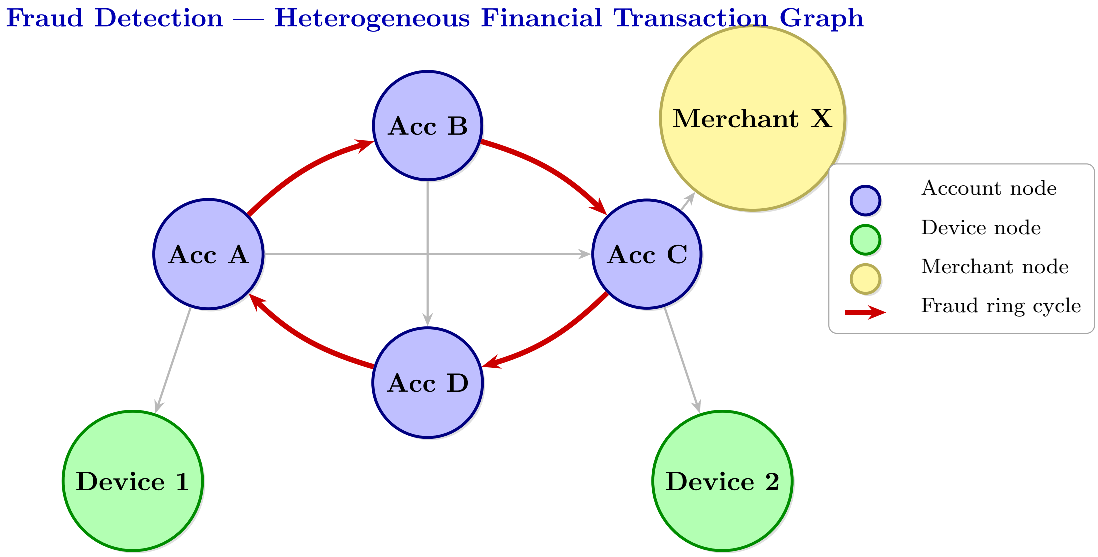

Figure 2: Heterogeneous financial graph: the red cycle (Acc A→B→C→D→A) reveals a fraud ring; shared device nodes expose colluding accounts.

2.3 Main Graph Concepts and Algorithms

Community detection (Louvain modularity) identifies dense subgraphs corresponding to fraud rings. Centrality measures—betweenness, eigenvector, PageRank—profile node importance correlated with fraud orchestrators. **GraphSAGE** [3] aggregates neighborhood features via trainable functions, producing node embeddings h_v for binary classification. It generalizes to unseen nodes—critical as new accounts are continuously created.

2.4 Why Classical Tabular ML Is Insufficient

Tabular classifiers treat each transaction in isolation. Fraud rings deliberately design individual transactions to appear normal. Representing shared infrastructure—two accounts on the same device—requires combinatorially expensive feature engineering, making it structurally intractable.

2.5 Strengths and Limitations

Strengths. Liu et al. [7] report precision-recall gains of up to 30% using GNN-based methods over tabular baselines. Fraud rings can be visualized as subgraphs, supporting regulatory reporting. **Limitations.** Extreme class imbalance (0.1–1% fraud rate) and billions of daily transactions require graph sampling and distributed processing.

3 Critical Reflection

Relational Data in Modern AI

The most challenging AI problems are those where meaning is *constituted by relationships.* Language, social interaction, biological function, and economic behavior are fundamentally relational. Attention mechanisms in transformers can be interpreted as learning dynamic graphs over input tokens—even non-graph architectures implicitly model relational structure, confirming it as a foundational AI requirement.

The Role of Graph Neural Networks

GNNs are the dominant paradigm combining structural inductive biases with learned representations. The message-passing framework [2]—where each node iteratively aggregates neighborhood information—subsumes many classical graph algorithms as special cases. Current frontiers include higher-order GNNs overcoming Weisfeiler-Leman expressivity limits, and GNN integration with large language models for grounded knowledge reasoning.

Scalability Challenges

Full-batch gradient computation is memory-intractable on large graphs; neighbor sampling [3] introduces approximation error accumulating with depth. Graph partitioning for distributed training causes cross-partition edge cuts that degrade quality. Streaming graphs require online update mechanisms incompatible with standard GNN pipelines—addressing these via sparsification and hardware-aware design remains one of the most active ML research areas.

References

[1] Bordes, A., et al. (2013). *Translating embeddings for modeling multi-relational data.* NeurIPS, 26, 2787–2795.

[2] Gilmer, J., et al. (2017). *Neural message passing for quantum chemistry.* ICML, 70, 1263–1272.

[3] Hamilton, W. L., Ying, R., & Leskovec, J. (2017). *Inductive representation learning on large graphs.* NeurIPS, 30, 1024–1034.

[4] Lao, N., & Cohen, W. W. (2010). *Relational retrieval using path-constrained random walks.* Machine Learning, 81(1), 53–67.

[5] Leblay, J., & Chekol, M. W. (2018). *Deriving validity time in knowledge graph.* WWW 2018, 1771–1776.

[6] Lewis, P., et al. (2020). *Retrieval-augmented generation for knowledge-intensive NLP tasks.* NeurIPS, 33, 9459–9474.

[7] Liu, Y., et al. (2021). *Pick and choose: A GNN-based imbalanced learning approach for fraud detection.* WWW 2021, 3168–3177.

[8] Veličković, P., et al. (2018). *Graph attention networks.* ICLR 2018. arXiv:1710.10903